

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/353751650>

Stream drying bioindication in Central Europe: A Biodrought Index accuracy assessment

Article in *Ecological Indicators* · August 2021

DOI: 10.1016/j.ecolind.2021.108045

CITATIONS

0

READS

51

9 authors, including:



[Michal Straka](#)

Masaryk University

29 PUBLICATIONS 128 CITATIONS

[SEE PROFILE](#)



[Marek Polášek](#)

Masaryk University

33 PUBLICATIONS 197 CITATIONS

[SEE PROFILE](#)



[Zoltán Csabai](#)

University of Pécs

160 PUBLICATIONS 1,233 CITATIONS

[SEE PROFILE](#)



[Oliver Zweidick](#)

University of Natural Resources and Life Sciences Vienna

9 PUBLICATIONS 0 CITATIONS

[SEE PROFILE](#)

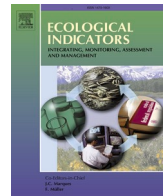
Some of the authors of this publication are also working on these related projects:



Karst ecology [View project](#)



Morávka river basin [View project](#)



Short Note

Stream drying bioindication in Central Europe: A Biodrought Index accuracy assessment

Michal Straka^{a,b,*}, Marek Polášek^{a,b}, Zoltán Csabai^{b,c}, Oliver Zweidick^d, Wolfram Graf^d, Elisabeth I. Meyer^e, Emília Mišíková Elexová^f, Margita Lešťáková^f, Petr Pařil^b

^a T. G. Masaryk Water Research Institute, p. r. i., Mojžírovo náměstí 16, 612 00 Brno, Czech Republic

^b Department of Botany and Zoology, Faculty of Science, Masaryk University, Kotlářská 2, 611 37 Brno, Czech Republic

^c Department of Hydrobiology, Faculty of Sciences, University of Pécs, Ifjúság útja 6, 7624 Pécs, Hungary

^d University of Natural Resources, Institute of Hydrobiology and Aquatic Ecosystem Management, Gregor-Mendel-Strasse 33, A-1180 Vienna, Austria

^e University of Münster, Institute for Evolution and Biodiversity, Hufferstr. 1, 48149 Münster, Germany

^f The Slovak National Water Reference Laboratory, Water Research Institute, Bratislava, Slovakia

ARTICLE INFO

Keywords:

Intermittent
Flow regime
Macroinvertebrates
Bioindication
Drought

ABSTRACT

With ongoing climate change and increasing water resource pressures, the knowledge and predictability of stream drying is essential for water management. However, the hydrological data for assessing the flow regime of temporary streams are often non-existent or scarce. The flow regime strongly affects stream ecological functioning and ecosystem processes. A few biological indices have been developed in the past decade to assess community responses to flow intermittency. One of them – the Biodrought Index, used to recognise antecedent stream drying based on benthic invertebrate assemblage composition – was initially intended for deployment in the Czech Republic. Here we report on a test of how reliably this index can be used in Central Europe. We used five independent data sets of macroinvertebrate samples collected during the flowing phase in five countries: Austria, Czech Republic, Germany, Hungary and Slovakia. Altogether 89, 26 and 220 samples were assessed from intermittent, near-perennial and perennial sites, respectively. Each sample was assigned to one of three flow intermittence categories according to the Biodrought Index and then compared with the flow regime recognised by hydrological observations and gauging station data. In most cases, the Biodrought Index clearly reflected the previous flow state and identified streamflow cessation. The misidentification rate between intermittent and perennial samples was very low, ranging from 0% to 6%. Classification rate of correctly recognised samples ranged from 54% to 93% and misclassification rate between near-perennial and perennial/intermittent category was 6% to 46%. Overall, this study confirmed the robustness and the reliability of the Biodrought Index, which proved to be an effective tool in assessing the recent hydrological history of small and medium-sized streams. The index can improve the interpretation of the macroinvertebrate assemblage data collected for ecological status assessment, can help to evaluate the effectiveness of river restoration projects or identify water resource pressures. Hence, we consider the Biodrought Index a useful method for indicating antecedent stream drying in the extensive area of Central Europe and thus helpful in monitoring and managing river ecosystems.

1. Introduction

Contrary to general assumptions, intermittent rivers and ephemeral streams (IRES) are not restricted exclusively to arid regions but occur in all biomes and ecoregions. The proportion of IRES in global river networks is likely to be greater than 50% (Datry et al., 2014). In humid climates (oceanic and continental), this proportion has increased continuously in recent decades, parallel with ongoing change in the

climate (Stahl et al., 2010; Laaha et al., 2017). The overall high significance of IRES for the global carbon cycle (Marcé et al., 2019), for unique and high biodiversity (Vander Vorste et al., 2019) and essential ecosystem services (Acuña et al., 2014; Datry et al., 2017) has been proved, while the recent onset of suprasedonal drought has raised awareness among public and expert communities (Van Lanen et al., 2016). Nevertheless, IRES are typically poorly represented in bio-monitoring programmes in Central European countries (Stubbington

* Corresponding author at: T. G. Masaryk Water Research Institute, p. r. i., Mojžírovo náměstí 16, 612 00 Brno, Czech Republic.

E-mail address: michal.straka@vuv.cz (M. Straka).

<https://doi.org/10.1016/j.ecolind.2021.108045>

Received 23 April 2021; Received in revised form 23 July 2021; Accepted 26 July 2021

Available online 6 August 2021

1470-160X/© 2021 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

et al., 2018). Our knowledge of the occurrence of flow intermittence is often limited because hydrological gauging station networks tend to cover only larger streams and rivers that are typically perennial in the temperate zone and have minimal risk of complete surface water loss. Thus, alternative methods to recognise zero-flow periods are greatly needed (Gallart et al., 2016; Beaufort et al., 2018; Fritz et al., 2020). To fill this gap, a few tools using benthic invertebrates have recently been developed (Cid et al., 2016; Chadd et al., 2017; England et al., 2019). Among them, the Biodrought Index (BI) (Straka et al., 2019) has been designed and tested for retrospective recognition of stream drying in the Czech Republic. Index calculation is based on analyses of benthic invertebrate samples collected within the stream's flowing phase (typically spring or autumn). It enables the calculation of the probability that the stream experienced flow cessation in the previous dry season. Its algorithm is season specific and is based on the proportion of indicator taxa, the proportion of taxa with high body flexibility, the proportion of taxa with a preference for organic substrates (autumn season) and total abundance (spring season). The original purpose of the BI was to evaluate benthic invertebrate samples that are routinely collected within the scope of monitoring programmes and to represent a new diagnostic tool that can improve evidence-driven stream management actions (Carvalho et al., 2019). In the Czech Republic, the index has shown a high accuracy of antecedent stream drying bioindication. However, the index's accuracy and reliability can be limited by factors like different sampling methods, taxonomic identification precision or varied species pools in study sites across climatic and geographical gradients. This work's overall aim is to test how well the BI performs outside of its range of development and possibly expand its geographical applicability. More specifically, we test the BI's reliability (i) in samples from five different

countries with different ecoregions/biogeographical regions, (ii) sampled with three different sampling techniques, and (iii) effect of taxonomic resolution. We hypothesise that BI efficiency will decrease in samples taken outside Czech Republic with different sampling methods and its reliability will decline with rougher taxonomic resolution.

2. Material and methods

2.1. Data sets

To test the BI's reliability, we compiled several data sets from benthic invertebrate studies across Central Europe. Only data sets of benthic invertebrate assemblages sampled from near-pristine small streams (Strahler orders 1–4) with a known recent hydrological history (i.e. extent of stream intermittence within a year before sampling) were used. Also, species/genus taxonomical resolution was demanded. Data sets from five countries were gathered and analysed: Austria, the Czech Republic, Germany, Hungary and Slovakia (Fig. 1, Table 1). The hydrological status of analysed streams was reconstructed using gauging station data, derived from in-stream temperature and water-level data-loggers as well as personal observations. The dry period duration allowed us to divide sampling sites into three categories: perennial, near-perennial and intermittent. Perennial sites were those with more than one year of continuous flow prior to the sampling campaign. Intermittent sites experienced complete surface water loss for weeks or months (greater than 7 days), and near-perennial sites were stated as an intermediate category when assignment to other two categories was uncertain. Typically the surface flow was lost for < 7 days, and series of disconnected pools were present. The only exception was the Czech data

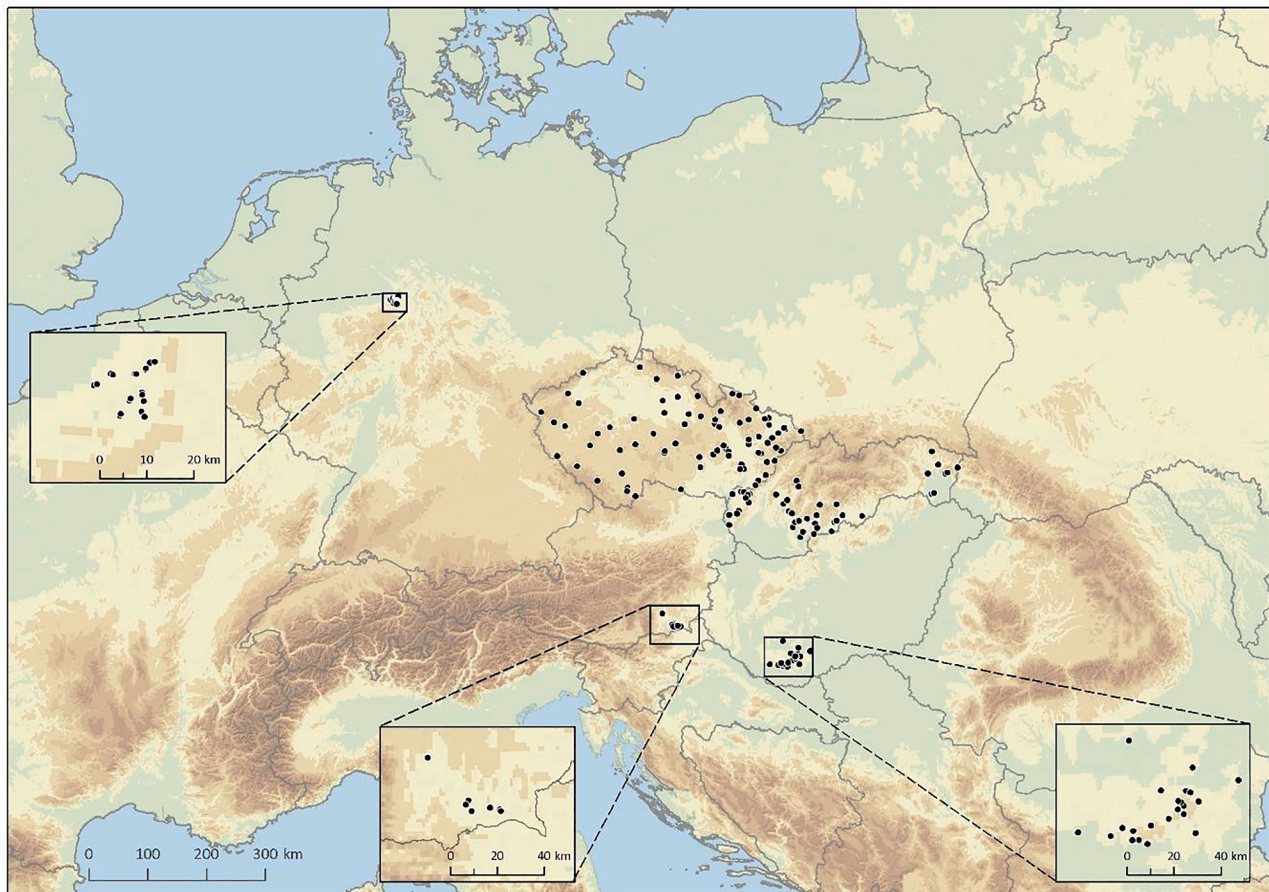


Fig. 1. Location of sites used for BI assessment. Twenty sites are located in East Westphalia (Germany), 76 sites are scattered across the Czech Republic, 8 sites are in Styria (south-eastern Austria), 22 sites are in the Mecsek mountains (southern Hungary) and 46 sites are located in Slovakia.

Table 1
Data sets used to explore the BI's effectiveness.

Country	Source	Number of samples			Hydrological monitoring	Sampling method	Overall taxonomic resolution [%]				Strahler's order	Altitude [m a.s.l.]	Ecoregion (sensu Illies, 1978)	Biogeographical region (EEA, 2016)
		Perennial	Near-perennial	Intermittent			Species	Genus	Family	Higher				
Austria (AT)	Zwiedick (2020)	2	3	3	visual observation; temperature dataloggers	modified AQEM; only five subsamples processed	48	12	19	21	1–2	250–340	Dinaric western Balkan	Continental
Czech Republic (CZ)	State monitoring data	117	26		visual observation	PERLA; three-minute multihabitat sampling	86	12	3	0	2–4	210–460	Central highlands, Hungarian lowlands	Continental, Pannonian
Germany (DE)	Meyer & Meyer (2000); Meyer et al. (2003)	29	3	39	visual observation	Surber; three Surber samples pooled together	75	10	15	0	2–4	140–380	Central plains	Continental
Hungary (HU)	Unpublished dataset	41	16	7	visual observation; temperature dataloggers; gauging stations	AQEM	90	4	6	0	1–3	125–380	Hungarian lowlands	Pannonian
Slovakia (SK)	State monitoring data	31	4	14	visual observation; gauging stations	AQEM	83	15	2	0	2–4	100–420	The Carpathians, Hungarian lowlands	Alpine, Pannonian

set, where it was not possible to distinguish between intermittent and near-perennial sites since no precise hydrological data were available. Hence, both categories were assessed together as intermittent.

Because of their different origins, the taxalists were heterogeneous concerning the identification level, sampling season and sampling and sorting methods. Three sampling methods were used during the flowing phase to collect benthic invertebrate samples: PERLA (Kokeš and Němejcová, 2006), AQEM (AQEM Consortium, 2002) and the use of a Surber sampler. While PERLA samples are time-related (three-minute kick sampling), the AQEM and Surber samples are area-related (Table 1). The data sets covered five different ecoregions as introduced by Illies (1978) and adopted in the European Union (EU) Water Framework Directive (European Commission, 2000), namely Dinaric Western Balkan, Central Highlands, Hungarian Lowlands, Central Plains and The Carpathians and three biogeographical regions according to the European Environment Agency (EEA, 2016), namely Continental, Pannonian and Alpine.

2.2. BI calculation and data analyses

The BI was calculated according to Straka et al. (2019). Before the calculation, each data set was adjusted to harmonise the taxalists with the Biodrought source database. This process comprises unification of taxa names, excluding non-relevant taxa (e.g. terrestrial species) and in case the taxon was missing in the database its adjustment to the nearest taxonomical unit.

The BI is season-specific and was originally designed for samples taken in spring or autumn. Given that the samples were taken during all seasons, we arbitrarily decided to assess winter (December, January, February) samples using the spring algorithm and summer (June, July, August) samples using an autumn-specific algorithm. According to the BI, which represents a discrimination score along an intermittence gradient, each sample was classified into one of the three flow intermittence categories (i.e. perennial, near-perennial or intermittent). This calculated classification was compared with the observed flow regime. The threshold values of the BI for classification to perennial/near-perennial and near-perennial/intermittent categories were adopted from Straka et al. (2019) and are 0.675 and −1.645, respectively. Flow permanence expressed as % of days in a year with continuous flow was studied within three data sets (Austria, Germany, Hungary) (Supplementary material 1). The relationship between flow permanence and the BI was examined using linear regression performed on available data from these three countries.

To test whether the BI can distinguish among the three intermittence categories, differences in index values among observed hydrological states were analysed using analysis of variance (ANOVA) for each country separately. The individual differences between categories' pairs were tested using planned contrasts using the R package lsmeans (Russell, 2016). In case of an insufficient number of replicates within a group, the post-hoc test was not performed. To test whether the original boundaries between intermittence categories calculated by Straka et al. (2019) are valid, we counted the percentage of correctly classified samples, the rate of samples misclassified to the closest category (intermittent/near-perennial and near-perennial/perennial) and the percentage of completely misidentified samples to the opposite category (intermittent/perennial and vice versa). In the Czech data set, only the misclassification of perennial/non-perennial was calculated because near-perennial and intermittent samples were not distinguished.

The probability density function of a normal distribution enables calculating the probability of classifying the analysed sample to each of the three intermittence categories (Straka et al. 2019). To test whether the samples' taxonomical identification level influences classification success, we fitted a linear model between the probability of classification to the correctly observed flow category and the identification level. The identification level was expressed as the percentage of macro-invertebrates identified as species. Additionally, we artificially

decreased the identification level from the original “best available” to “genus” and “family” levels, respectively. Afterwards, the BI values were calculated and compared among observed flow categories for all three identification levels separately.

3. Results

The BI was calculated for 335 samples from five evaluated data sets. Overall, 220 perennial, 26 near-perennial and 89 intermittent samples were assessed. The index values ranged from -7.4 (sample from a Czech intermittent stream) to 3.9 (sample from a Hungarian perennial stream). We found highly significant differences in BI values among the observed sample categories in each country (Fig. 2). Regarding pairwise differences, we found significant differences between BI values of opposite categories (intermittent and perennial) and also intermittent and near-perennial types. In the case of the Austrian dataset, the difference between perennial and other categories was not tested since the number of replicates in perennial samples was low ($N = 2$). Differences in the BI between near-perennial and perennial types were significant only in the dataset from Germany (Fig. 2.). However, the index values for the near-perennial category were transitional between the intermittent and the perennial categories in all countries and the BI reflects continuous changes in stream flow duration (Supplementary material 1).

Each sample was classified into one of three categories according to the index. This classification was compared with the known flow regime history, and the success rate of the correct classification was calculated. The correct classification of samples varied from 54% in Hungary to 93% in Germany and the Czech Republic (Fig. 3, Supplementary material 2). The rate of complete misidentification (i.e. intermittent sample misclassified as perennial and vice versa) was very low. Overall, only six of 335 samples were misclassified between intermittent and perennial samples.

The identification level varied considerably among the assessed

samples, ranging from samples with 100% species level identification to samples in which more than 50% of invertebrates were identified to family or a higher taxonomic level. However, overall taxonomic resolution was mainly to species or genus level (Table 1) and the probability of classification to the correct category was not affected by the sample identification level (ANOVA $F_{1,328} = 1.54$, $p = 0.22$). Artificially decreased identification level was reflected by decreasing in BI efficiency (Supplementary material 3).

4. Discussion and conclusions

Our study demonstrated the generalizability of a new biomonitoring tool, the BI, to data sets collected in different European ecoregions with various sampling methods and diverse taxonomic resolutions. A wide range of approaches exists for biomonitoring using macroinvertebrates across Europe (Friberg et al., 2006). This lack of methodological uniformity emphasises the importance of comparing the effectiveness of bioassessment indices across different regions (Herbst and Silldorff, 2006). Furthermore, sample processing in a laboratory is known to be a possible reason for bias in environmental assessments (Haase et al., 2010; Ligeiro et al., 2020). Therefore, we gathered independent data sets (collected and analysed by different teams applying differing sampling and processing protocols) to evaluate the potential of the index for future use across a wide geographical range. The BI was originally developed to assess samples taken with the PERLA method; we added another two common sampling methods (AQEM and Surber sampler), processed within different laboratories and according to their own laboratory protocols. The results showed that the BI can effectively distinguish intermittent and perennial samples even though different sampling methods were used (Fig. 3). This finding is in concordance with Lorenz and Clarke (2006), who found time-related PERLA samples to be highly comparable with area-related AQEM samples.

The BI is a multimetric index based on the responses of specific taxa

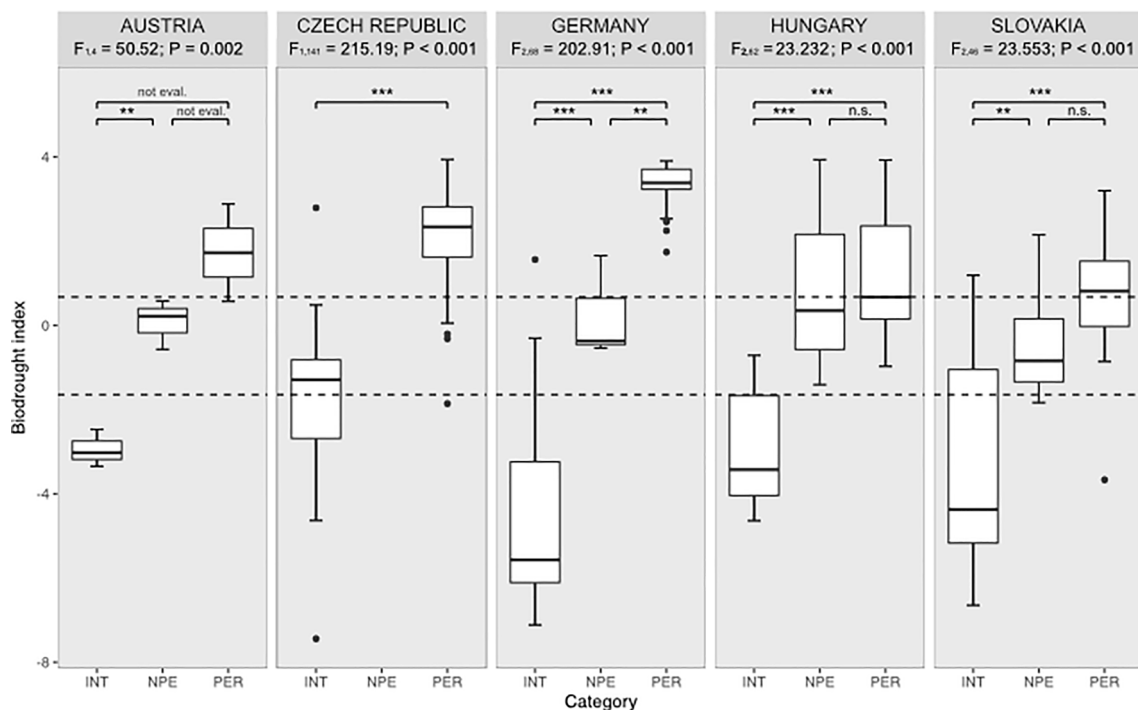


Fig. 2. Distribution of the values of the BI calculated for samples from five countries. Individual boxplots represent the classification of samples observed according to hydrological data (INT = intermittent, NPE = near-perennial, PER = perennial). The higher/lower the index values, the more perennial/intermittent the samples (according to the index). Dashed lines represent the thresholds between the intermittent/near-perennial and near-perennial/perennial categories according to Straka et al. (2019). The F statistic and the P values for ANOVA comparing all three intermittence categories are shown in each country plot's header; asterisks represent the significance of the pairwise differences tested using planned contrasts (*** = $p < 0.001$; ** = $p < 0.01$; * = $p < 0.1$; n.s. = $p > 0.1$; not eval. = groups with insufficient number of replicates were not statistically tested).

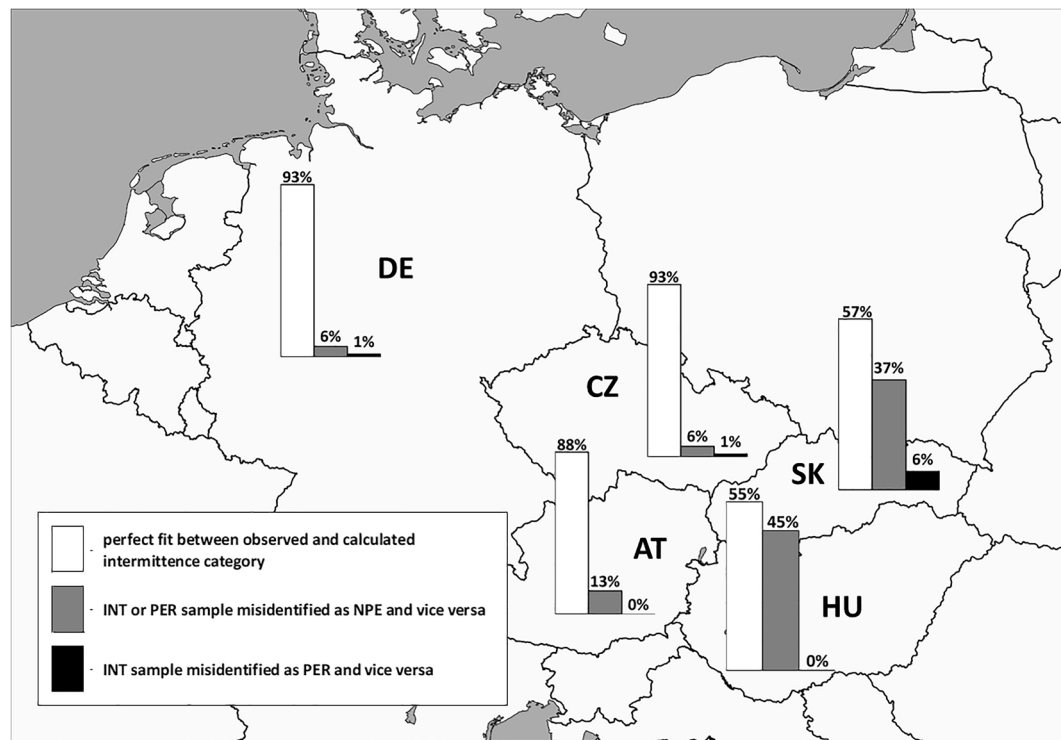


Fig. 3. Percentage of classification of samples according to the BI compared to the observed intermittence categories. White columns represent correctly classified samples. Grey columns represent misidentifications to the neighbouring category (i.e. perennial or intermittent classified as near-perennial and vice versa). Black columns represent complete misidentifications (i.e. perennial sample classified as intermittent and vice versa).

to stream drying (proportion of indicator taxa), assemblage parameters (total abundance) and species traits proportion (body flexibility, preference for organic substrates) (Straka et al., 2019). Metrics based on specific species traits are comparable across geographic regions, as evidenced by applying indices such as LIFE index (Lotic-invertebrate Index for Flow Evaluation) in hydroecological modelling across and beyond Europe (Dunbar et al., 2010; Ncube et al., 2018). The one of the crucial BI metrics is a ratio of flow intermittence sensitive and flow intermittence tolerant taxa. The recognition of these taxa was based on data from the Czech Republic. The similarity of species composition between ecological communities was found to decrease with increasing geographical distance (Astorga et al., 2012). Consequently the use of a taxa-based index can be limited outside the area where it was developed. Thus, it is challenging to determine whether indices developed for small areas can be extended beyond the areas, for which they were initially designed for (Stoddard et al., 2008; Martins et al., 2020). According to the results presented in this study, we can conclude that the geographical limits of the index were not reached. The samples from CZ, DE and AT were classified into intermittence categories with similar success rates, and the BI can be used outside area where it was constructed. However, the high proportion of samples misidentified within the near-perennial category in the Hungarian and Slovakian data sets suggests that wider use of the BI must be carefully assessed. As these data sets origin from the eastern part of the study area it could indicate that invertebrate assemblages during the dry phase form different mechanisms in oceanic-continental climate gradient. This was suggested by Paril et al. (2019) who recorded high proportion of alive aquatic invertebrates without drought-resistance adaptations in dry stream sediments in continental climate in contrast to previously reported findings from oceanic climates. High similarity of Hungarian perennial and near-perennial samples may be caused by past dry events since severe drought can affect assemblages for years (Bogan and Lytle, 2011) and our study included hydrological status only one year back. These issues remain to be tested at a more extensive time scale and a larger

geographic area than the one tested here.

Macroinvertebrate taxonomic resolution can influence the performance of hydrologically sensitive indices (Monk et al., 2012) and considerable bias in index performance has already been recognised for the Saprobic (Sandin and Hering, 2004) or LIFE index (Laini et al., 2018) when a higher taxonomic identification level (family level or above) is used. We did not observe the decreasing reliability of sample classification with reduced taxonomic resolution in the original data sets. However, this might have been due to the fact that even in the data set with the least detailed taxonomic resolution, 60% of individuals were identified to the species or genus level. When the identification level was artificially decreased, the BI reliability deteriorated. Therefore, we consider the fine taxonomic resolution to be a fundamental prerequisite of correct BI performance.

Overall, the results confirmed that biomonitoring tools can indicate flow intermittency (Sarremejane et al., 2019; White et al., 2019). The benthic community metrics reflected the gradient of drying severity (Sarremejane et al., 2020) and the near-perennial category showed intermediate values between intermittent and perennial sites (Fig. 2). Stream drying is a gradual process when available habitats are reduced and community composition and diversity change (Lake, 2003). After the flow resumes, the consecutive rate and the extent of community recovery is influenced by the proximity of perennial refuges (White et al., 2018; Doretto et al., 2019), the length and the severity of drought (Datry, 2012; Bogan et al., 2015), the time since flow resumption (Calapez et al., 2014; Doretto et al., 2019) the sediment character (Vadher et al., 2018; Loskotová et al., 2019) and the site's history (Crabot et al., 2020). These factors can greatly influence community recovery and possibly mask the effect of previous stream drying. The response of the BI can be obscured and its performance under such conditions (e.g. distance to refuges) remains to be tested.

In conclusion, we believe that the BI is an effective index especially to identify streams with continuous flow and streams that have previously experienced long-term (weeks to months) flow cessation. The

samples categorised as an intermediate near-perennial category need to be assessed as “suspicious” and the flow regime assessment requires further information (e.g. hydrological data, temperature loggers, visual observation). BI thus represents a diagnostic tool with sufficient capability to identify if a stream drying had occurred in the one-year period preceding sample collection in all studied countries (Austria, Czech Republic, Germany, Hungary and Slovakia). However, the BI's reliability outside of tested conditions range (i.e. in polluted, acidified, morphologically modified streams or streams above 500 m a.s.l.) and outside of tested geographical area (e.g. use in Mediterranean, Western Europe) is currently unknown.

CRediT authorship contribution statement

Michal Straka: Conceptualization, Funding acquisition, Visualization, Writing - original draft. **Marek Polášek:** Data curation, Formal analysis, Methodology, Visualization, Writing - review & editing. **Zoltán Csabai:** Data curation, Investigation, Writing - review & editing. **Oliver Zweidick:** Data curation, Investigation, Writing - review & editing. **Wolfram Graf:** Data curation, Investigation, Writing - review & editing. **Elisabeth I. Meyer:** Data curation, Investigation, Writing - review & editing. **Emília Misíková Elexová:** Data curation, Investigation, Writing - review & editing. **Margita Lešáková:** Data curation, Investigation, Writing - review & editing. **Petr Paříl:** Conceptualization, Funding acquisition, Investigation, Methodology, Supervision, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

This work was supported by Long-term development conception of research organisation VÚV TGM, p. r. i. M. Straka, M. Polášek and P. Paříl were supported by the Czech Science Foundation (P505-20-17305S) and the INTER-COST project INTER-EXCELLENCE (MEYS LTC17017). Z. Csabai was supported by the grants 20765-3/2018/FEKUTSTRAT, TUDFO/47138/2019-ITM and NKFIH FK135136. We thank Ondra Hájek for creating the map and reviewers for valuable comments which improve final version of this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ecolind.2021.108045>.

References

- Acuña, V., Datry, T., Marshall, J., Barcelo, D., Dahm, C.N., Ginebreda, A., McGregor, G., Sabater, S., Tockner, K., Palmer, M.A., 2014. Why should we care about temporary waterways? *Science* 343 (6175), 1080–1081. <https://doi.org/10.1126/science.1246666>.
- AQEM Consortium, 2002. Manual for the application of the AQEM system. A comprehensive method to assess European streams using benthic macroinvertebrates, developed for the purpose of the Water Framework Directive. Version 1.0 (www.aqem.de), February (2002).
- Astorga, A., Oksanen, J., Luotko, M., Soininen, J., Virtanen, R., Muotka, T., 2012. Distance decay of similarity in freshwater communities: do macro- and microorganisms follow the same rules? *Glob. Ecol. Biogeogr.* 21, 365–375. <https://doi.org/10.1111/j.1466-8238.2011.00681.x>.
- Beaufort, A., Lamouroux, N., Pella, H., Datry, T., Sauquet, E., 2018. Extrapolating regional probability of drying of headwater streams using discrete observations and gauging networks. *Hydrol. Earth Syst. Sci.* 22: 3033–3051. <https://www.hydrol-earth-syst-sci.net/22/3033/2018/>.
- Bogan, M.T., Boersma, K.S., Lytle, D.A., 2015. Resistance and resilience of invertebrate communities to seasonal and suprasedasonal drought in arid-land headwater streams.

- Freshw. Biol. 60 (12), 2547–2558. <https://doi.org/10.1111/fwb.2015.60.issue-1210.1111/fwb.12522>.
- Bogan, M.T., Lytle, D.A., 2011. Severe drought drives novel community trajectories in desert stream pools. *Freshw. Biol.* 56, 2070–2081. <https://doi.org/10.1111/j.1365-2427.2011.02638.x>.
- Calapez, A.R., Elias, C.L., Almeida, S.F.P., Feio, M.J., 2014. Extreme drought effects and recovery patterns in the benthic communities of temperate streams. *Limnetica* 33, 281–296. <https://doi.org/10.23818/limn.33.22>.
- Carvalho, L., Mackay, E.B., Cardoso, A.C., Baattrup-Pedersen, A., Birk, S., Blackstock, K.L., Boriss, G., Borja, A., Feld, C.K., Ferreira, M.T., Globevnik, L., Grizzetti, B., Hendry, S., Hering, D., Kelly, M., Langaas, S., Meissner, K., Panagopoulos, Y., Penning, E., Rouillard, J., Sabater, S., Schmedtje, U., Spears, B.M., Venohr, M., van de Bund, W., Solheim, A.L., 2019. Protecting and restoring Europe's waters: an analyses of the future development needs of the Water Framework Directive. *Sci. Total Environ.* 658, 1228–1238. <https://doi.org/10.1016/j.scitotenv.2018.12.255>.
- Cid, N., Verkaik, I., García-Roger, E.M., Rieradevall, M., Bonada, N., Sánchez-Montoya, M.M., Gómez, R., Suárez, M.L., Vidal-Abarca, M.R., Demartini, D., Buffagni, A., Erba, S., Karaouzias, I., Skoulikidis, N., Prat, N., 2016. A biological tool to assess flow connectivity in reference temporary streams from the Mediterranean Basin. *Sci. Total Environ.* 540, 178–190. <https://doi.org/10.1016/j.scitotenv.2015.06.086>.
- Crabot, J., Polášek, M., Launay, B., Paříl, P., Datry, T., 2020. Drying in newly intermittent rivers leads to higher variability of invertebrate communities. *Freshw. Biol.* 66 (4), 730–744. <https://doi.org/10.1111/fwb.v66.410.1111/fwb.13673>.
- Datry, T., 2012. Benthic and hyporheic invertebrate assemblages along a flow intermittence gradient: effects of duration of dry events. *Freshw. Biol.* 57, 563–574. <https://doi.org/10.1111/j.1365-2427.2011.02725.x>.
- Datry, T., Larned, S.T., Tockner, K., 2014. Intermittent rivers: a challenge for freshwater ecology. *Bioscience* 64, 229–235. <https://doi.org/10.1093/biosci/bit027>.
- Datry, T., Boulton, A.J., Bonada, N., Fritz, K., Leigh, C., Sauquet, E., Tockner, K., Hugueny, B., Dahm, C.N., Vamasi, S., 2017. Flow intermittence and ecosystem services in rivers of the Anthropocene. *J. Appl. Ecol.* 55 (1), 353–364. <https://doi.org/10.1111/jpe.2018.55.issue-110.1111/1365-2664.12941>.
- Doretto, A., Bona, F., Falasco, E., Morandini, D., Piano, E., Fenoglio, S., 2019. Stay with the flow: How macroinvertebrate communities recover during the rewetting phase in Alpine streams affected by an exceptional drought. *River Res. Appl.* 36 (1), 91–101. <https://doi.org/10.1002/rra.v36.110.1002/rra.3563>.
- Dunbar, M.J., Pedersen, M.L., Cadman, D., Extence, C.H., Waddingham, J., Chadd, R., Larsen, S.E., 2010. River discharge and local-scale physical habitat influence macroinvertebrate LIFE scores. *Freshw. Biol.* 55, 226–242. <https://doi.org/10.1111/j.1365-2427.2009.02306.x>.
- England, J., Chadd, R., Dunbar, M.J., Sarremejane, R., Stubbington, R., Westwood, C.G., Leeming, D., 2019. An invertebrate-based index to characterize ecological responses to flow intermittence in rivers. *Fund. Appl. Limnol.* 193 (1), 93–117. <https://doi.org/10.1127/fal/2019/1206>.
- European Commission, 2000. Directive 2000/60/EC of the European Parliament of the Council of 23rd October 2000 establishing a framework for community action in the field of water policy. *Official J. Eur. Commun.* 327, 1–72.
- EEA, 2016. European Environment Agency. Biogeographical Regions. Available from <https://www.eea.europa.eu/data-and-maps/data/biogeographical-regions-europe-3>.
- Friberg, N., Sandin, L., Furse, M.T., Larsen, S.E., Clarke, R.T., Haase, P., 2006. Comparison of macroinvertebrate sampling methods in Europe. *Hydrobiologia* 566 (1), 365–378. <https://doi.org/10.1007/s10750-006-0083-6>.
- Fritz, K.M., Nadeau, T.-L., Kelso, J.E., Beck, W.S., Mazor, R.D., Harrington, R.A., Topping, B.J., 2020. Classifying streamflow duration: the scientific basis and an operational framework for method development. *Water* 12, 2545. <https://doi.org/10.3390/w12092545>.
- Gallart, F., Llorens, P., Latron, J., Cid, N., Rieradevall, M., Prat, N., 2016. Validating alternative methodologies to estimate the regime of temporary rivers when flow data are unavailable. *Sci. Total Environ.* 565, 1001–1010. <https://doi.org/10.1016/j.scitotenv.2016.05.116>.
- Haase, P., Pauls, S.U., Schindehütte, K., Sundermann, A., 2010. First audit of macroinvertebrate samples from an EU Water Framework Directive monitoring program: human error greatly lowers precision of assessment results. *J. North Am. Benthol. Soc.* 29 (4), 1279–1291. <https://doi.org/10.1899/09-183.1>.
- Herbst, D.B., Silldorff, E.L., 2006. Comparison of the performance of different bioassessment methods: similar evaluations of biotic integrity from separate programs and procedures. *J. North Am. Benthol. Soc.* 25, 513–530. [https://doi.org/10.1899/0887-3593\(2006\)25\[513:COTPOD\]2.0.CO;2](https://doi.org/10.1899/0887-3593(2006)25[513:COTPOD]2.0.CO;2).
- Chadd, R.P., England, J.A., Constable, D., Dunbar, M.J., Extence, C.A., Leeming, D.J., Murray-Bligh, J.A., Wood, P.J., 2017. An index to track the ecological effects of drought development and recovery on riverine invertebrate communities. *Ecol. Ind.* 82, 344–356. <https://doi.org/10.1016/j.ecolind.2017.06.058>.
- Illies, J., 1978. *Limnofauna Europaea. A Checklist of the Animals Inhabiting European Inland Waters, with an Account of their Distribution and Ecology, 2nd Edition.* Gustav Fischer Verlag, Stuttgart, p. 552.
- Kokeš, J., Němejcová, D., 2006. Metodika odběru a zpracování vzorků makrozoobentosu tekoucích vod metodou PERLA [Methodology for macrozoobenthos sampling in running waters by PERLA method]. Metodika VÚV TGM Praha.
- Laaha, G., Gauster, T., Tallaksen, L.M., Vidal, J.P., Stahl, K., Prudhomme, C.H., Heudorfer, B., Vlnas, R., Ionita, M., Van Lanen, H.A.J., Adler, M.-J., Caillouet, L., Delus, C., Fendekova, M., Gailliez, S., Hannaford, J., Kingston, D., Van Loon, A.F., Mediero, L., Osuch, M., Romanowicz, R., Sauquet, E., Stagge, J.H., Wong, W.K., 2017. The European 2015 drought from a hydrological perspective. *Hydrol. Earth Syst. Sci.* 21, 3001–3024. <https://doi.org/10.5194/hess-21-3001-2017>.

- Laini, A., Bolpagni, R., Cancellario, T., Guareschi, S., Racchetti, E., Viaroli, P., 2018. Testing the response of macroinvertebrate communities and biomonitoring indices under multiple stressors in a lowland regulated river. *Ecol. Ind.* 90, 47–53. <https://doi.org/10.1016/j.ecolind.2018.02.051>.
- Lake, P.S., 2003. Ecological effects of perturbation by drought in flowing waters. *Freshw. Biol.* 48, 1161–1172. <https://doi.org/10.1046/j.1365-2427.2003.01086.x>.
- Ligeiro, R., Hughes, R.M., Kaufmann, P.R., Heino, J., Melo, A.S., Callisto, M., 2020. Choice of field and laboratory methods affects the detection of anthropogenic disturbances using stream macroinvertebrate assemblages. *Ecol. Ind.* 115, 106382. <https://doi.org/10.1016/j.ecolind.2020.106382>.
- Lorenz, A., Clarke, R.T., 2006. Sample coherence – a field study approach to assess similarity of macroinvertebrate samples. *Hydrobiologia* 566, 461–476. https://doi.org/10.1007/978-1-4020-5493-8_32.
- Loskotová, B., Straka, M., Pařil, P., 2019. Sediment characteristics influence benthic macroinvertebrate vertical migrations and survival under experimental water loss conditions. *Fund. Appl. Limnol.* 193 (1), 39–49. <https://doi.org/10.1127/fal/2019/1138>.
- Marcé, R., Obrador, B., Gómez-Gener, L., Catalán, N., Koschorreck, M., Arce, M.A., Singer, G., von Schiller, D., 2019. Emissions from dry inland waters are a blind spot in the global carbon cycle. *Earth Sci. Rev.* 188, 240–248. <https://doi.org/10.1016/j.earscirev.2018.11.012>.
- Martins, I., Macedo, D.R., Hughes, R.M., Callisto, M., 2020. Are multiple multimetric indices effective for assessing ecological condition in tropical basin? *Ecol. Ind.* 110, 105953. <https://doi.org/10.1016/j.ecolind.2019.105953>.
- Meyer, A., Meyer, E.L., 2000. Discharge regime and the effect of drying on macroinvertebrate communities in a temporary karst stream in East Westphalia (Germany). *Aquat. Sci.* 62 (3), 216–231. <https://doi.org/10.1007/PL00001333>.
- Meyer, A., Meyer, E.L., Meyer, C., 2003. Lotic communities of two small temporary karstic stream systems (East Westphalia, Germany) along a longitudinal gradient of hydrological intermittency. *Limnologia* 33 (4), 271–279. [https://doi.org/10.1016/S0075-9511\(03\)80022-1](https://doi.org/10.1016/S0075-9511(03)80022-1).
- Monk, W.A., Wood, P.J., Hannah, D.M., Extence, C.A., Chadd, R.P., Dunbar, M.J., 2012. How does macroinvertebrate taxonomic resolution influence ecohydrological relationships in riverine ecosystems. *Ecohydrology* 5 (1), 36–45. <https://doi.org/10.1002/eco.v5.110.1002/eco.192>.
- Ncube, S., Visser, A., Beevers, L.C., 2018. A framework for assessing instream supporting ecosystem services based on hydroecological modelling. *Water* 10, 1247. <https://doi.org/10.3390/w10091247>.
- Pařil, P., Polásek, M., Loskotová, B., Straka, M., Crabot, J., Datry, T., 2019. An unexpected source of invertebrate community recovery in intermittent streams from a humid continental climate. *Freshw. Biol.* 64 (11), 1971–1983. <https://doi.org/10.1111/fwb.13386>.
- Russell V. Lenth, 2016. Least-squares means: the R package lsmeans. *J. Statist. Softw.* 69 (1): 1–33. <https://doi.org/10.18637/jss.v069.i01>.
- Sandin, L., Hering, D., 2004. Comparing Macroinvertebrate Indices to Detect Organic Pollution across Europe: A Contribution to the EC Water Framework Directive Intercalibration. In: Hering, D., Verdonschot, P.F.M., Moog, O., Sandin, L. (Eds.), *Integrated Assessment of Running Waters in Europe. Developments in Hydrobiology*, vol 175. Springer, Dordrecht. https://doi.org/10.1007/978-94-007-0993-5_4.
- Sarremejane, R., Stubbington, R., Dunbar, M.J., Westwood, C.G., England, J., 2019. Biological indices to characterize community responses to drying in streams with contrasting flow permanence regimes. *Ecol. Ind.* 107, 105620. <https://doi.org/10.1016/j.ecolind.2019.105620>.
- Sarremejane, R., England, J., Sefton, C.E.M., Parry, S., Eastman, M., Stubbington, R., 2020. Local and regional drivers influence how aquatic community diversity, resistance and resilience vary in response to drying. *Oikos* 129 (12), 1877–1890. <https://doi.org/10.1111/oik.12911>.
- Stahl, K., Hisdal, H., Hannaford, J., Tallaksen, L.M., van Lanen, H.A.J., Sauquet, E., Demuth, S., Fendekova, M., Jódar, J., 2010. Streamflow trends in Europe: evidence from a dataset of near-natural catchments. *Hydrol. Earth Syst. Sci.* 14, 2367–2382. <https://doi.org/10.5194/hess-14-2367-2010>.
- Stoddard, J.L., Herlihy, A.T., Peck, D.V., Hughes, R.M., Whittier, T.R., Tarquinio, E., 2008. A process for creating multimetric indices for large-scale aquatic surveys. *Freshw. Sci.* 27 (4), 878–891. <https://doi.org/10.1899/08-053.1>.
- Straka, M., Polásek, M., Syrovátka, V., Stubbington, R., Zahrádková, S., Němejcová, D., Šikulová, L., Rezníčková, P., Opatřilová, L., Datry, T., Pařil, P., 2019. Recognition of stream drying based on benthic macroinvertebrates: a new tool in Central Europe. *Ecol. Ind.* 106, 105486. <https://doi.org/10.1016/j.ecolind.2019.105486>.
- Stubbington, R., Chadd, R., Cid, N., Csabai, Z., Miliša, M., Morais, M., Munné, A., Pařil, P., Pešić, V., Tziortzis, I., Verdonschot, R.C.M., Datry, T., 2018. Biomonitoring of intermittent rivers and ephemeral streams in Europe: current practice and priorities to enhance ecological status assessments. *Sci. Total Environ.* 618, 1096–1113. <https://doi.org/10.1016/j.scitotenv.2017.09.137>.
- Vadher, A.N., Millett, J., Stubbington, R., Wood, P.J., 2018. Drying duration and stream characteristics influence macroinvertebrate survivorship within the sediments of a temporary channel and exposed gravel bars of a connected perennial stream. *Hydrobiologia* 814 (1), 121–132. <https://doi.org/10.1007/s10750-018-3544-9>.
- Vander Vorste, R., Sarremejane, R., Datry, T., 2019. Intermittent rivers and ephemeral streams: A unique biome with important contributions to biodiversity and ecosystem services. Reference module in Earth Systems and Environmental Sciences. Elsevier. doi: 10.1016/B978-0-12-409548-9.12054-8.
- Van Lanen, H.A.J., Laaha, G., Kingston, D.G., Gauster, T., Ionita, M., Vidal, J.-P., Vlnas, R., Tallaksen, L.M., Stahl, K., Hannaford, J., Delus, C., Fendekova, M., Mediero, L., Prudhomme, C., Rets, E., Romanowicz, R.J., Gailliez, S., Wong, W.K., Adler, M.-J., Blauhut, V., Caillouet, L., Chelcea, S., Frolova, N., Gudmundsson, L., Hanel, M., Haslinger, K., Kireeva, M., Osuch, M., Sauquet, E., Stagge, J.H., Van Loon, A.F., 2016. Hydrology needed to manage droughts: the 2015 European case. *Hydrol. Process.* 30 (17), 3097–3104. <https://doi.org/10.1002/hyp.v30.1710.1002/hyp.10838>.
- White, J.C., House, A., Punchedard, N., Hannah, D.M., Wilding, N.A., Wood, P.J., 2018. Macroinvertebrate community responses to hydrological controls and groundwater abstraction effects across intermittent and perennial headwater streams. *Sci. Total Environ.* 310–611, 1514–1526. <https://doi.org/10.1016/j.scitotenv.2017.06.081>.
- White, J.C., Armitage, P.D., Bass, J.A.B., Chadd, R.P., Hill, M.J., Mathers, K.L., Little, S., Wood, P.J., 2019. How freshwater biomonitoring tools vary sub-seasonally reflects temporary river flow regimes. *River Res. Appl.* 35, 1325–1337. <https://doi.org/10.1002/rra.3501>.
- Zweidick, O., 2020. Macroinvertebrate communities of intermittent streams in Bioregion 14 (“Grazer Feld und Grabenland”, Austria: Styria) with focus on caddisflies. University of Natural Resources and Life Sciences, Vienna, Master thesis, Graz, 125pp.